EX.NO:4 CREATE AN ABSTRACT CLASS – SHAPE CLASS

PROGRAM:

```
import java.util.Scanner;

abstract class LibraryMember {
    int memberId;
    String name, email, phone;

    LibraryMember(int memberId, String name, String email, String phone) {
        this.memberId = memberId;
        this.name = name;
        this.email = email;
        this.phone = phone;
    }

    void display() {
        System.out.println("Member ID : " + memberId);
        System.out.println("Name      : " + name);
        System.out.println("Email     : " + email);
        System.out.println("Phone     : " + phone);
    }

    abstract void generateSummary();
}

class StudentMember extends LibraryMember {

    StudentMember(int id, String name, String email, String phone) {
        super(id, name, email, phone);
    }

    void generateSummary() {
        System.out.println("\n----- Student Member -----");
        display();
        System.out.println("Borrowing Limit      : 5 Books");
        System.out.println("Penalty Per Day      : Rs.2");
        System.out.println("Annual Membership Fee: Rs.200");
    }
}

class FacultyMember extends LibraryMember {

    FacultyMember(int id, String name, String email, String phone) {
        super(id, name, email, phone);
    }

    void generateSummary() {
        System.out.println("\n----- Faculty Member -----");
        display();
        System.out.println("Borrowing Limit      : 10 Books");
        System.out.println("Penalty Per Day      : Rs.3");
        System.out.println("Annual Membership Fee: Rs.500");
    }
}

class ExternalMember extends LibraryMember {
    ExternalMember(int id, String name, String email, String phone) {
```

```
        super(id, name, email, phone);
    }

    void generateSummary() {
        System.out.println("\n----- External Member -----");
        display();
        System.out.println("Borrowing Limit      : 3 Books");
        System.out.println("Penalty Per Day      : Rs.5");
        System.out.println("Annual Membership Fee: Rs.1000");
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Member ID: ");
        int id = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Email: ");
        String email = sc.nextLine();
        System.out.print("Enter Phone: ");
        String phone = sc.nextLine();
        System.out.println("\nSelect Member Type");
        System.out.println("1. Student");
        System.out.println("2. Faculty");
        System.out.println("3. External");
        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        switch (choice) {
            case 1:
                StudentMember s = new StudentMember(id, name, email, phone);
                s.generateSummary();
                break;
            case 2:
                FacultyMember f = new FacultyMember(id, name, email, phone);
                f.generateSummary();
                break;
            case 3:
                ExternalMember e = new ExternalMember(id, name, email, phone);
                e.generateSummary();
                break;
            default:
                System.out.println("Invalid Choice");
        }
        sc.close();
    }
}
```

OUTPUT:

Sample Run 1 (Student)

```
Enter Member ID: 101
Enter Name: Janaki
Enter Email: janaki@gmail.com
Enter Phone: 9876543210

Select Member Type
1. Student
2. Faculty
3. External
Enter Choice: 1

----- Student Member -----
Member ID : 101
Name      : Janaki
Email     : janaki@gmail.com
Phone     : 9876543210
Borrowing Limit      : 5 Books
Penalty Per Day      : Rs.2
Annual Membership Fee: Rs.200
```

Sample Run 2 (Faculty)

```
Enter Member ID: 101
Enter Name: Janaki
Enter Email: janaki@gmail.com
Enter Phone: 9876543210

Select Member Type
1. Student
2. Faculty
3. External
Enter Choice: 2

----- Faculty Member -----
Member ID : 101
Name      : Janaki
Email     : janaki@gmail.com
Phone     : 9876543210
Borrowing Limit      : 10 Books
Penalty Per Day      : Rs.3
Annual Membership Fee: Rs.500
```

Sample Run 3 (External)

```
Enter Member ID: 101
Enter Name: Janaki
Enter Email: janaki@gmail.com
Enter Phone: 9876543210

Select Member Type
1. Student
2. Faculty
3. External
Enter Choice: 3

----- External Member -----
Member ID : 101
Name      : Janaki
```

```
Email     : janaki@gmail.com
Phone     : 9876543210
Borrowing Limit      : 3 Books
Penalty Per Day      : Rs.5
Annual Membership Fee: Rs.1000
```